5.4 Solving Special Systems of Linear Equations

Name _____ Period ____ Date _____

Algebra I 5.4 (30-minute plan) · Student sheet

I can…

- tell how many solutions a system has.
- solve systems with no solution or infinitely many solutions.
- use special systems to solve real-life problems.

Vocabulary · fill in the missing word

No solution — _____ lines (same slope, different y-intercepts); solving gives a false statement such as 0 = 5

Infinitely many solutions — the same line (same slope and _____); solving gives a true statement such as 0 = 0

One solution — _____ slopes

Part 1 · Notes and examples (fill in each note, then work the example)

Parallel lines: Same slope, different y-intercepts: _____ solution (a _____ statement).

Example 1

Solve.

$2x + y = 4$

$4x + 2y = 3$

Different slopes: Different slopes: _____ solution.

Same line: Same slope and y-intercept: _____ solutions (a _____ statement).

Example 2

Solve.

$x - 3y = 6$

$-2x + 6y = -12$

Part 2 · Partner practice

Partner practice

How many solutions? $y = 4x - 2$ and $8x - 2y = 4$